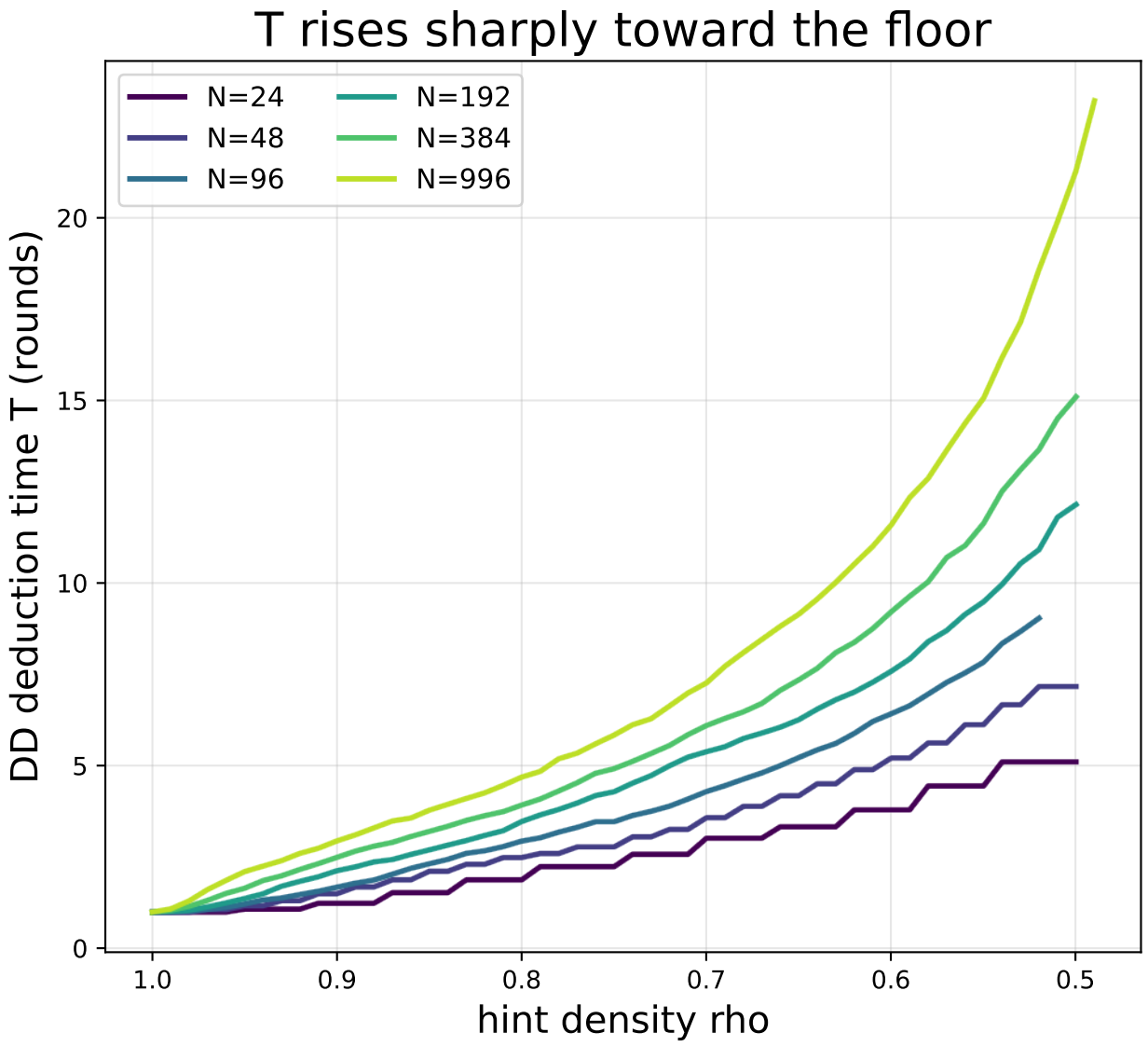
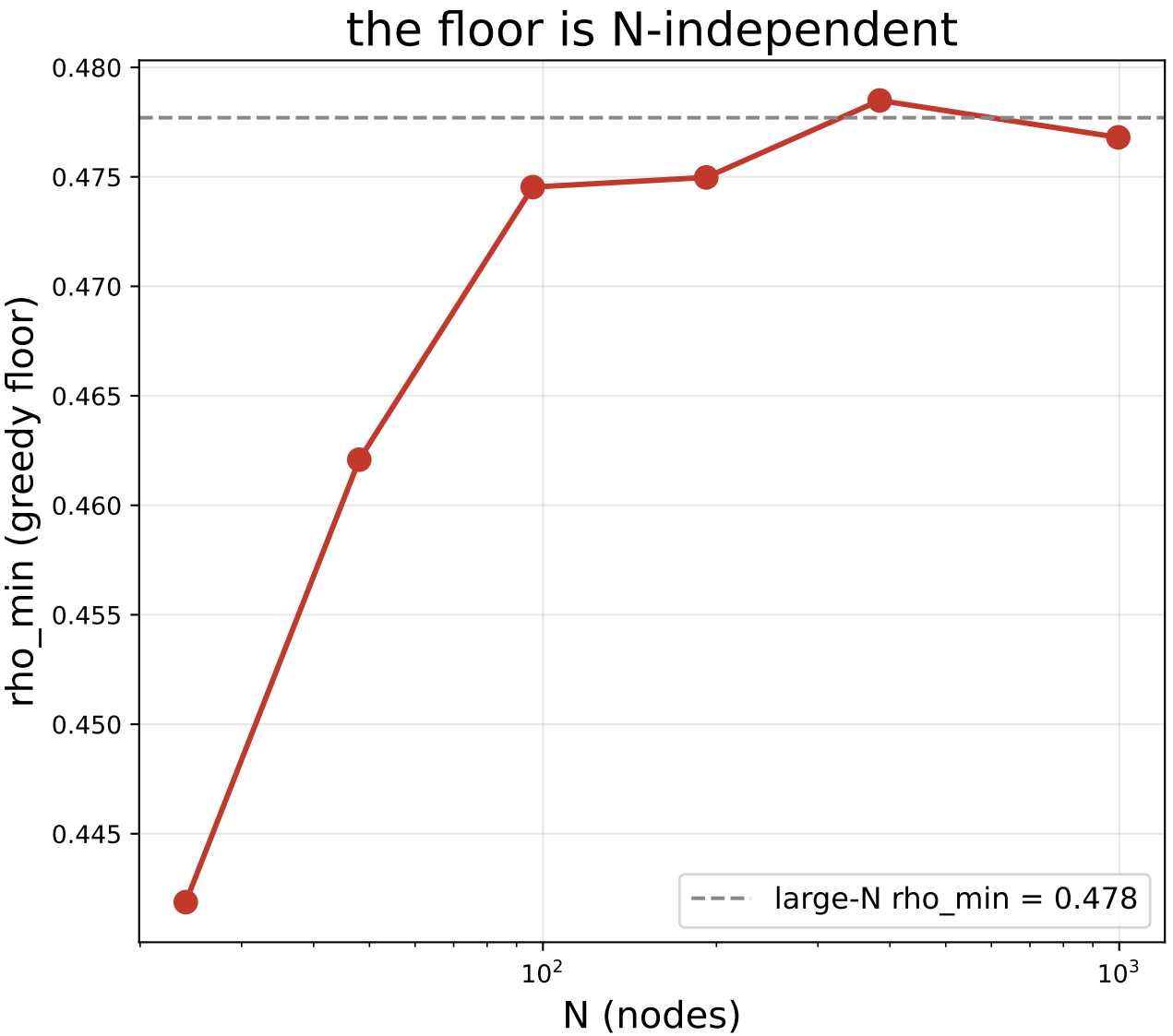
DD-Sudoku transformer

hint-density transition + what algorithm it uses

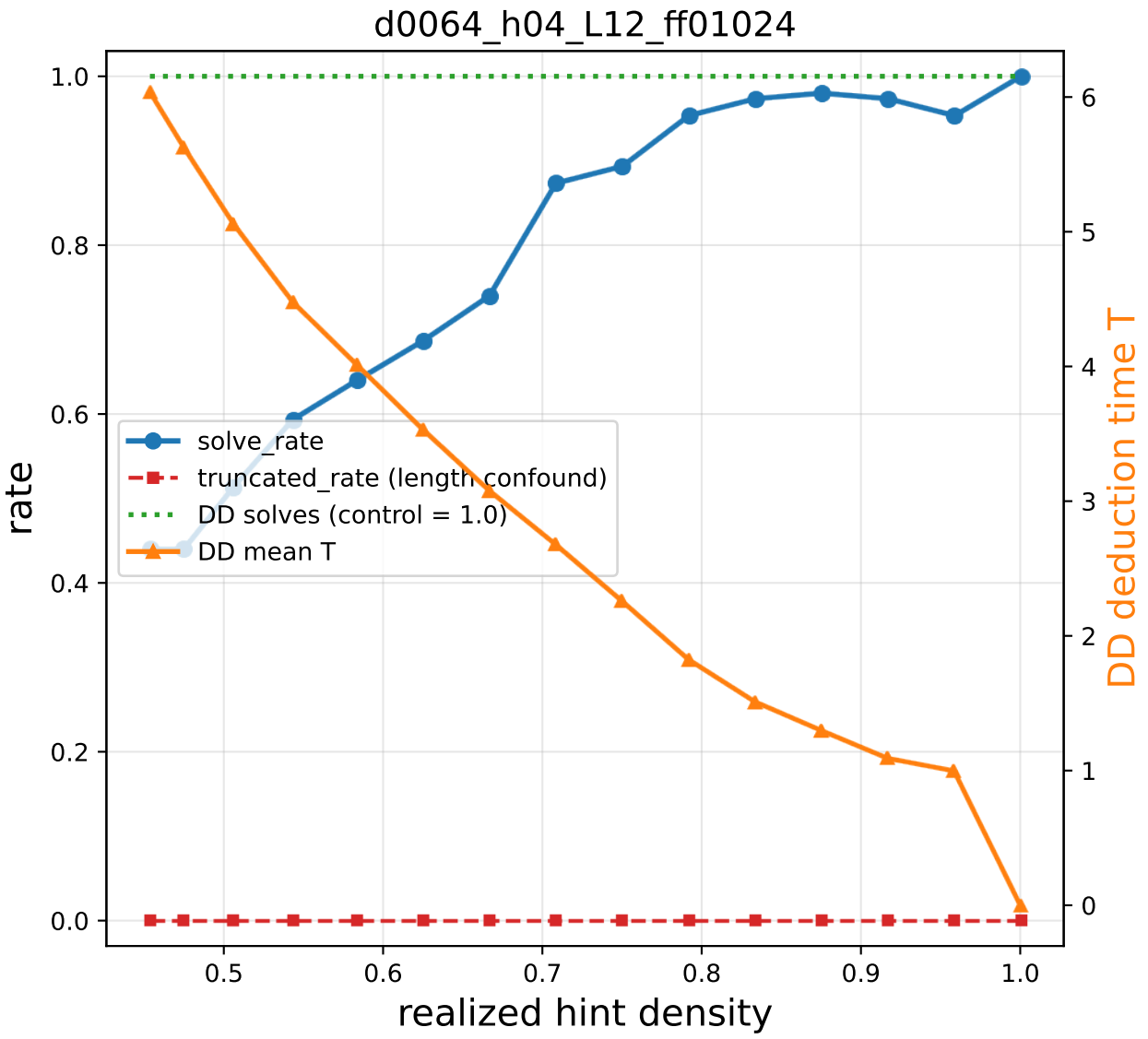
The transition IS there: solve_rate 0.99 -> 0.44 as DD's deduction depth goes 1 -> 6.
But the model is not running DD -- it tracks which nodes are colored (known_mask 0.92) rather than which are deducible (frontier_mask 0.67, control 0.57), and loses the head-to-head vote against a sorted-order heuristic.

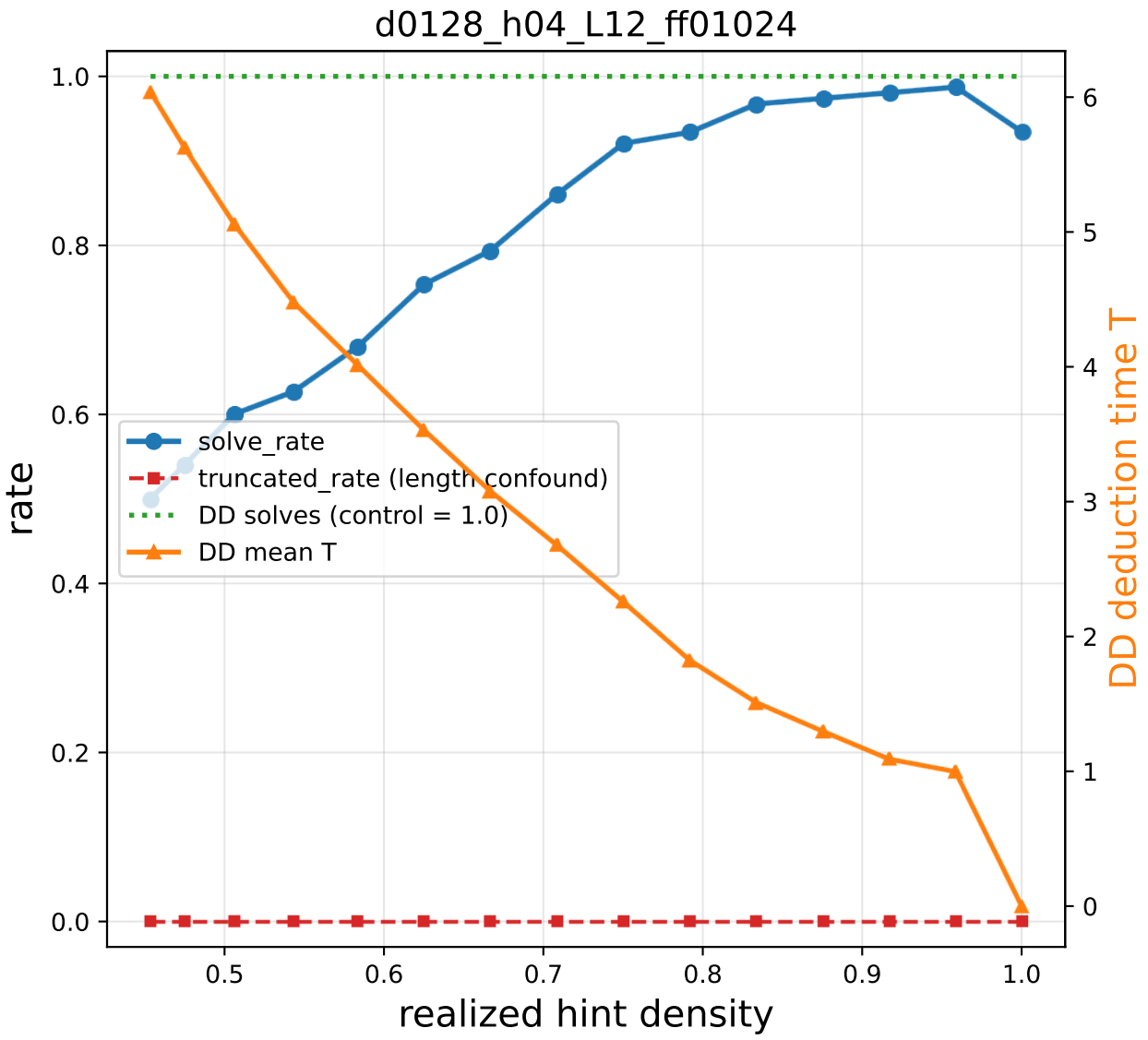
Where the transition is: DD deduction time diverges at rho_min



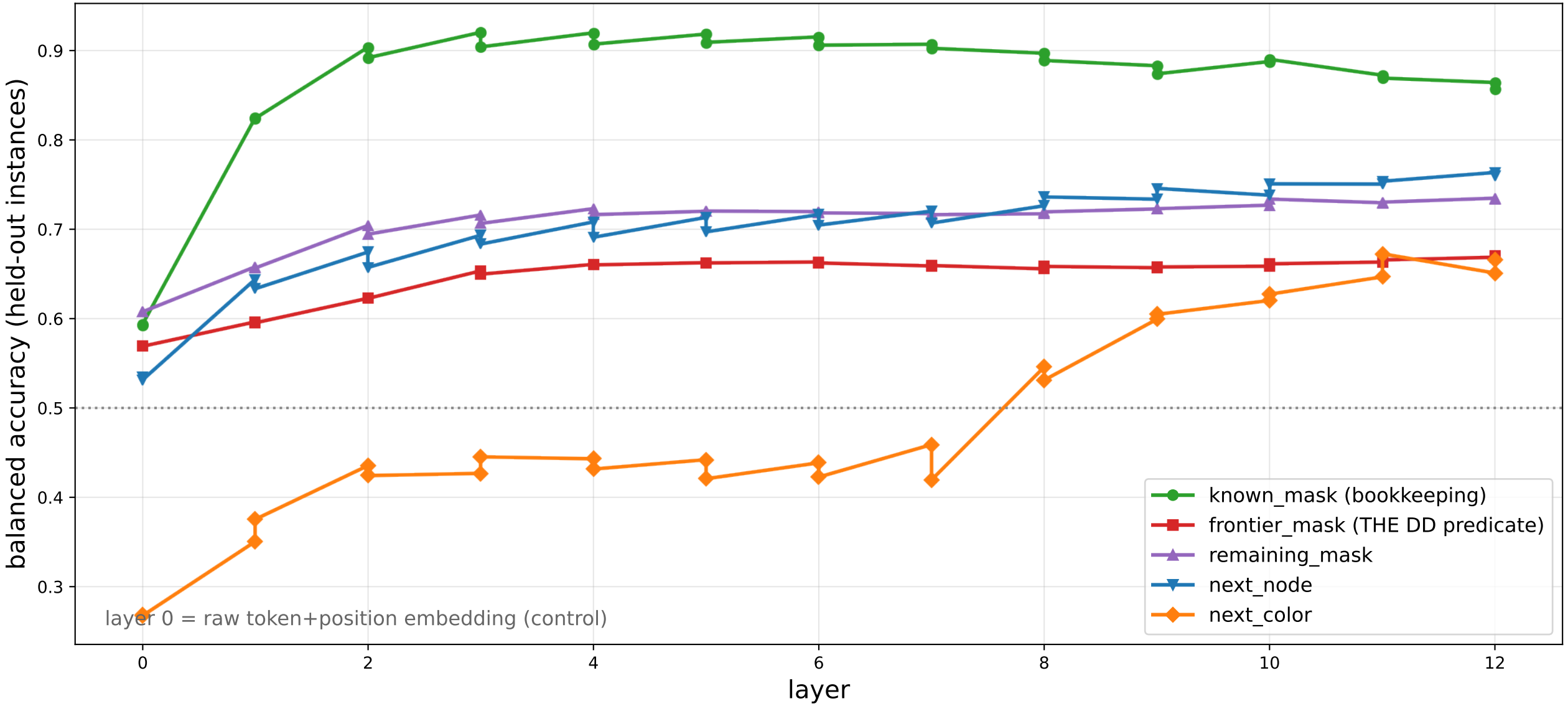


Does the model collapse where DD's reasoning depth diverges?

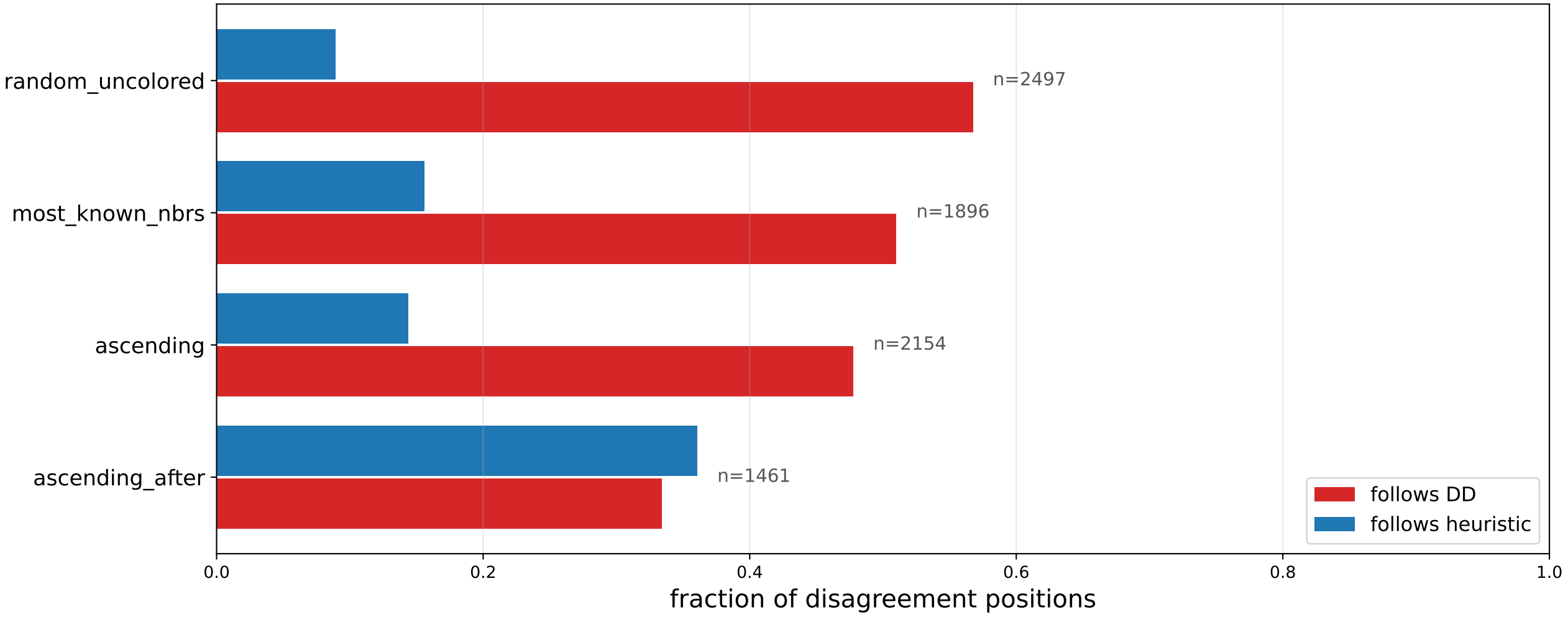




Probes: bookkeeping yes, DD predicate no

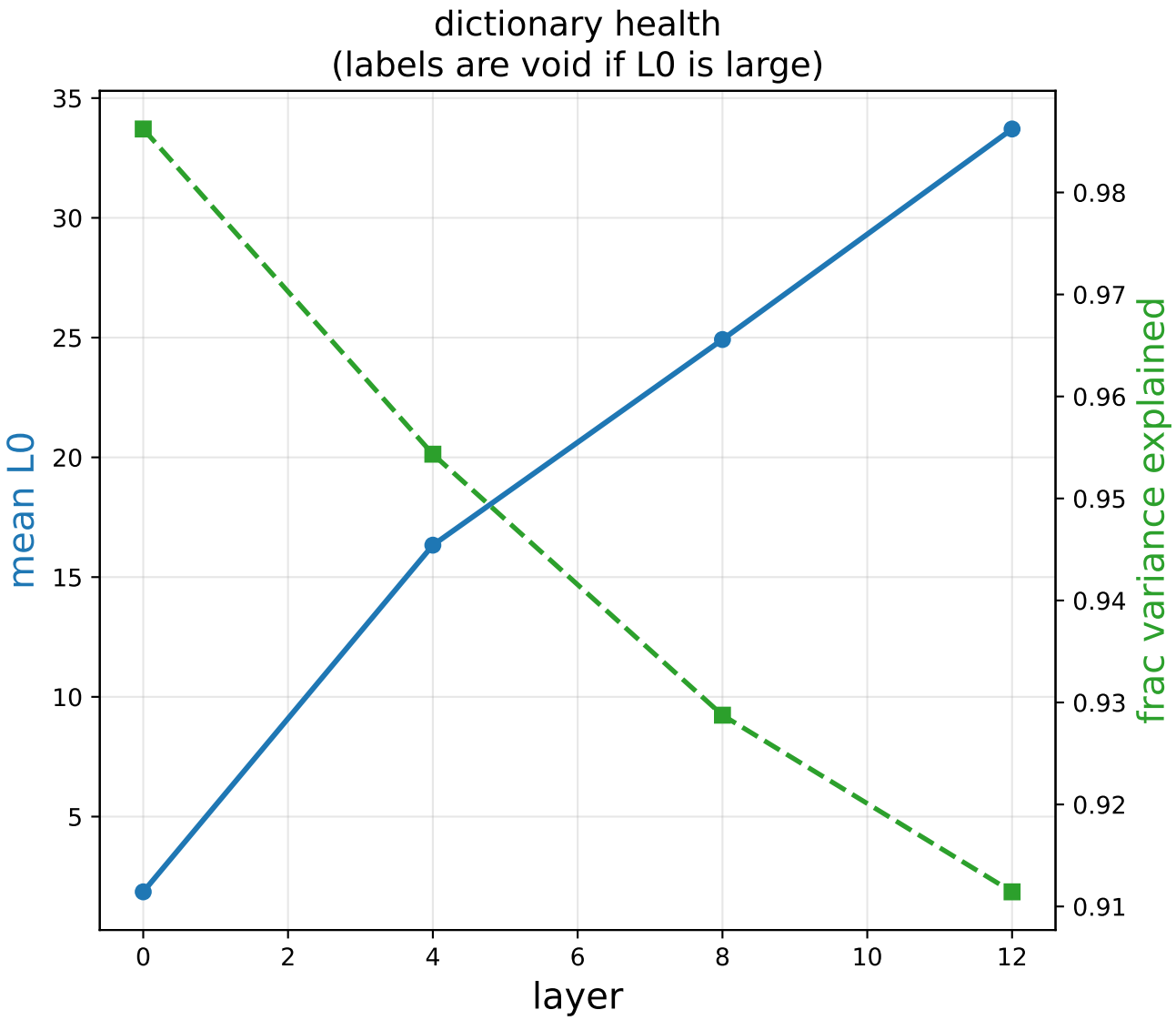


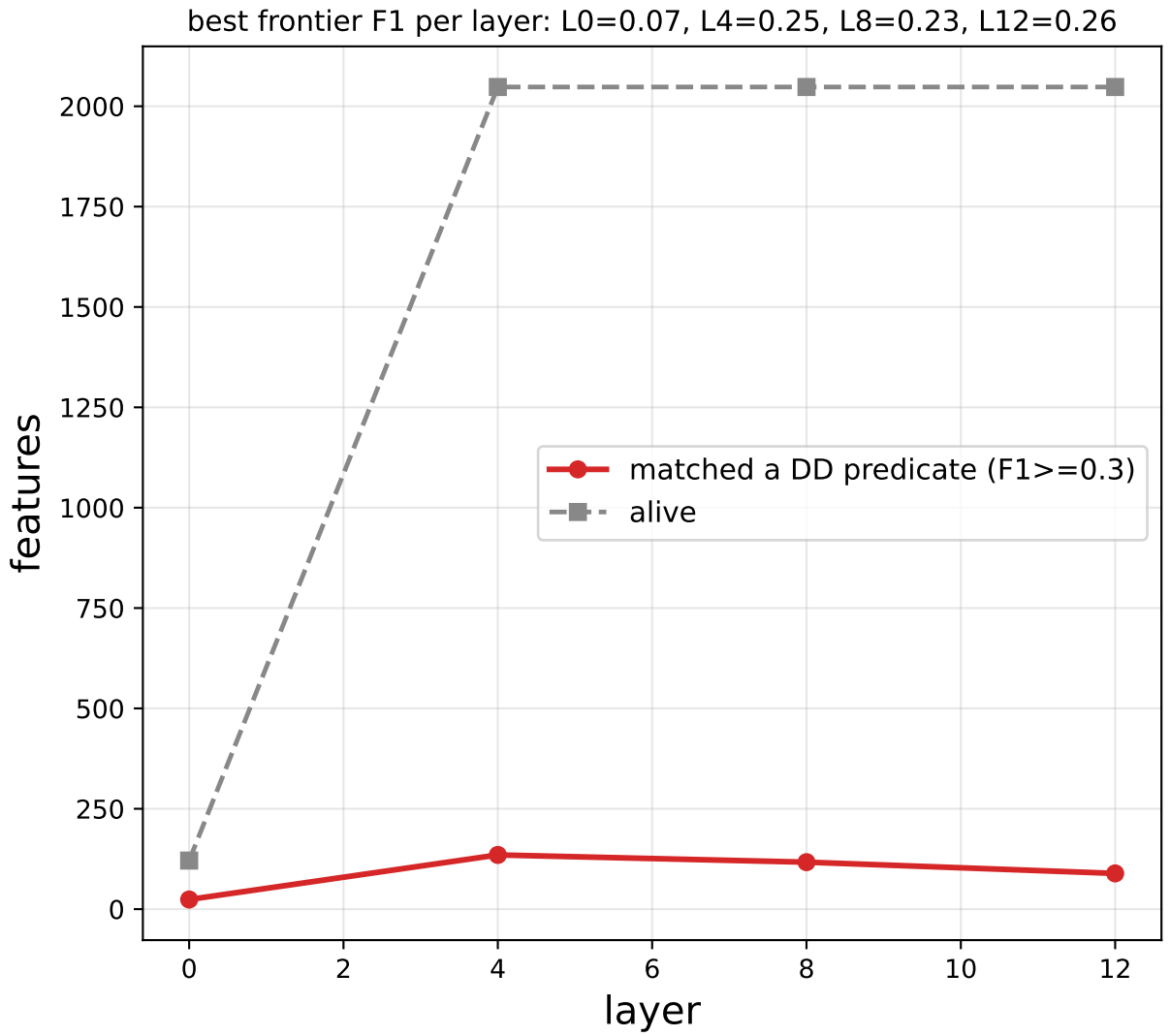
Which rule wins where DD and a heuristic disagree?



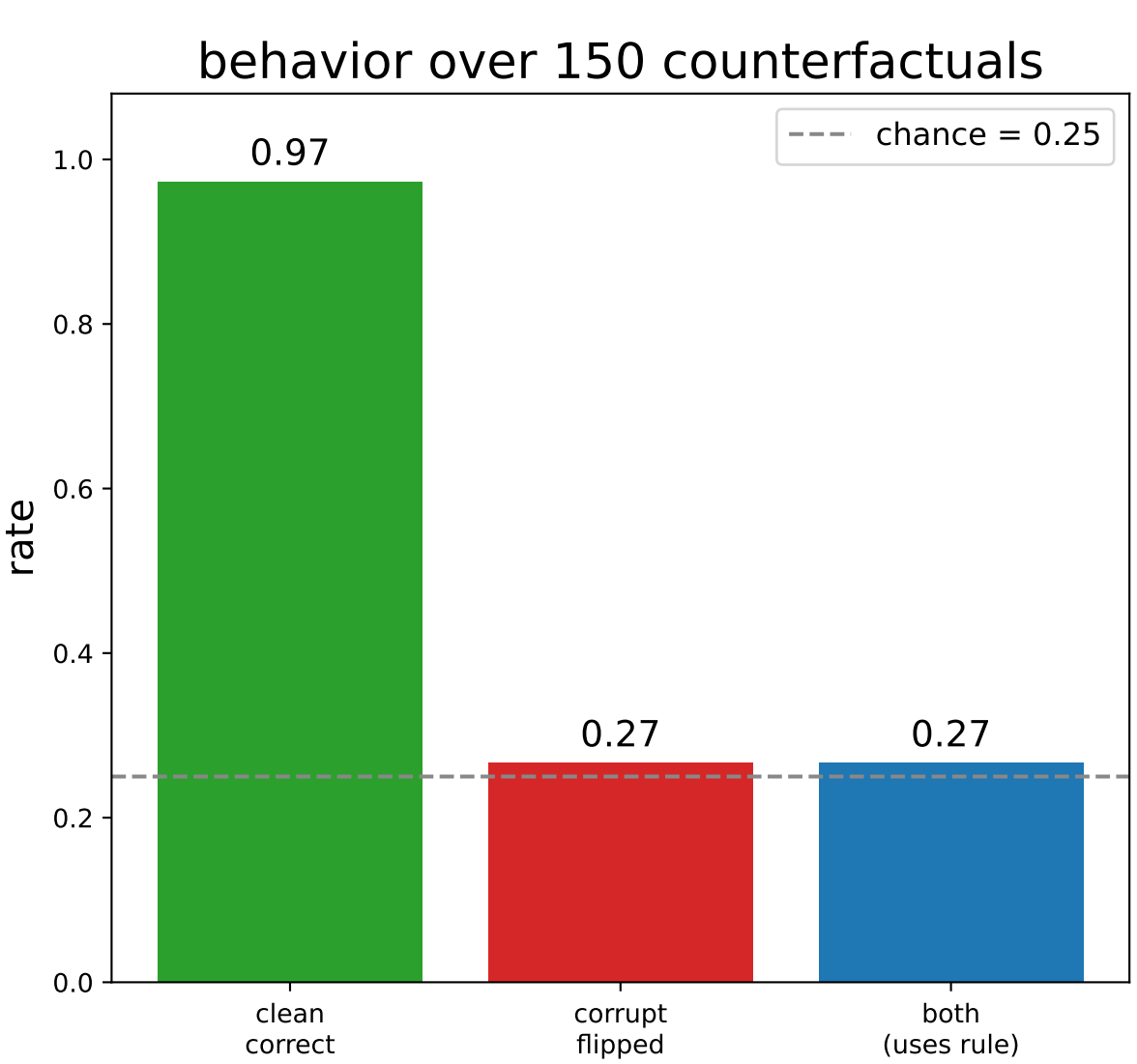
ascending_after = 'next uncolored id above the last emitted, wrapping'
the only baseline that exploits the sorted-emission convention -- and it beats DD

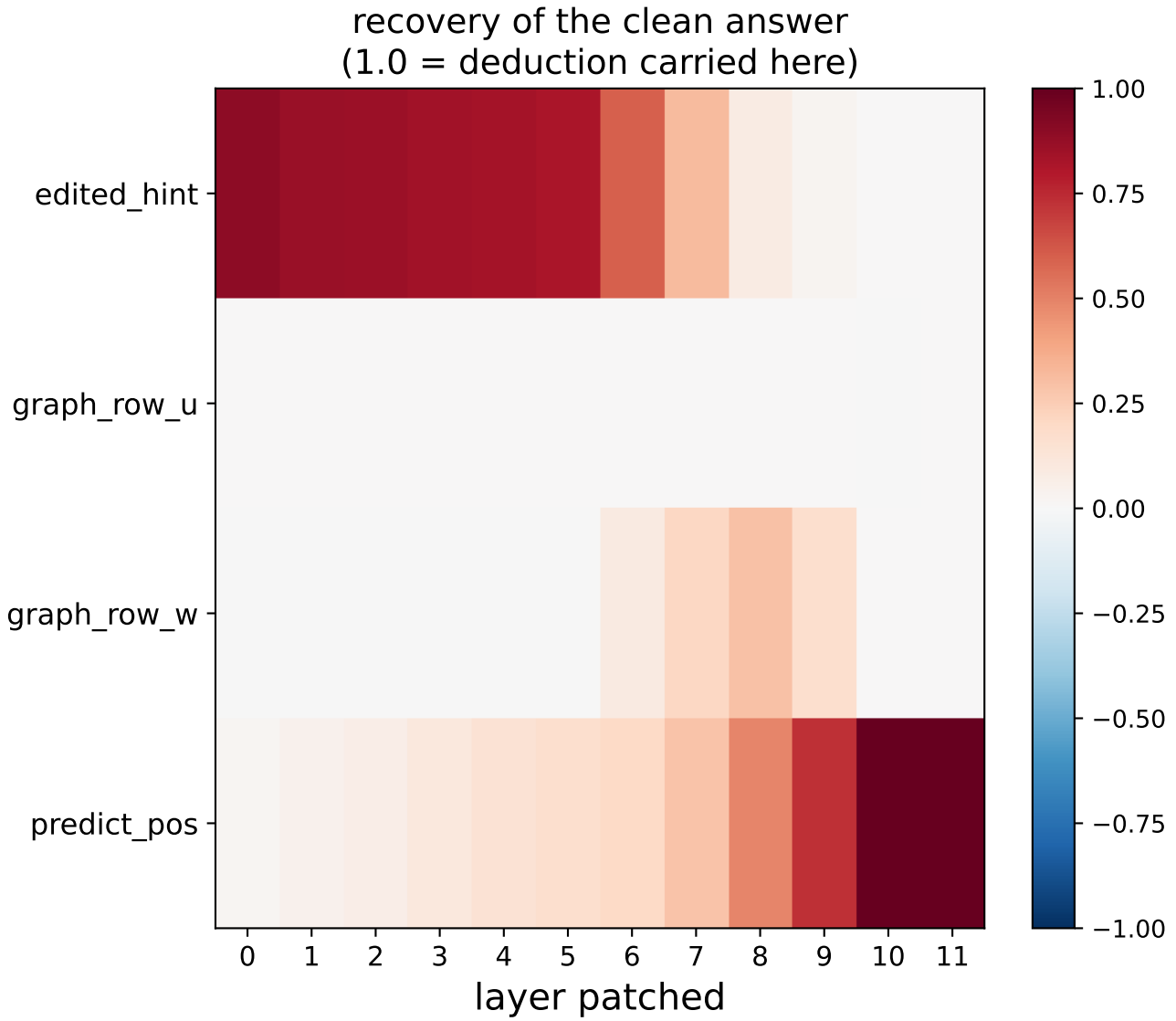
SAE: do sparse directions individually mean DD predicates?





Causal test: one neighbour-color token flips DD's answer





When during training does DD appear?

not available yet:
/scratch/users/hshunt/spin_gnn/sudoku_series_probes_2026-10-19/series_probes.csv

Summary and what it means

1. The hint-density transition is NOT in independent random hints -- that ensemble is subcritical at every density, with recovery identical from N=24 to N=384. It is in the removal-path ensemble the models already train on, as a divergence of DD's deduction time T as rho approaches the floor rho_min = 0.478 (N-independent).

2. The gradient exists at N=24, so the transition is testable on existing checkpoints. This retires the earlier plan to scale N to ~1700 tokens. MEASURED: solve_rate falls 0.99 -> 0.44 monotonically as DD's depth rises T=1.0 -> 6.0, with truncated_rate = 0.000 in every bin, so this is reasoning depth and not generation budget.

3. Probes: bookkeeping (known_mask, 0.92) is computed early and strongly; DD's deducibility predicate (frontier_mask, 0.67 vs a 0.57 control) is barely present. next_color is computed late (0.46 at L7 -> 0.65 at L12).

4. Causal test: the model does NOT apply the forcing rule at decision level (flip rate 0.27 vs 0.25 chance), but neighbour-color information does flow: the corruption moves the logit gap by 12.3. Patching localizes that flow to edited_hint (L0-5) -> graph row of the RECOLORED NEIGHBOUR (L6-9) -> prediction position (L8-11). The row of the node being predicted carries ~0.00 throughout, so this is color-to-node binding, not DD's neighbour-list join.

5. Honest gaps: head ablation in this run was a no-op (fixed, rerunning as 48585025); 'when during training' awaits the checkpoint series; and the scratchpad variant came in at agg_solve_rate 0.740 vs the 0.767 baseline, i.e. no improvement.